

CIDR (Classless Inter-Domain Routing) is a flexible way to allocate IP addresses and route internet traffic. Introduced in 1993, it replaced the old, rigid classful network system (classes A, B, and C) to prevent the world from running out of IPv4 addresses.

Think of CIDR as a system where you can buy custom-sized pizza slices, whereas the old system forced you to buy an entire pizza even if you only wanted one bite.

The problem: why we needed CIDR

In the early days of the internet, IP addresses were given out in fixed, inflexible blocks called classes.

- o class A: offered 167 million IPs (Given to massive telecom giants or governments).

- o class B: offered 65,536 IPs. (Given to universities or medium/large companies).

- o class C: offered 256 IPs (Given to small business).

Real-world Analogy of the problem.

Imagine a company needs 2000 IP addresses for its employees.

- A Class C Block (256 IPs) is too small.
- So the registry had to give them a class B Block (65,536 IPs),
- Result: The company used 2000 IPs and wasted 63,536 IPs! nobody else could use those remaining addresses.

This waste caused IPv4 addresses to deplete rapidly, CIDR was created to solve this exact issue by allowing networks to be sized to any variable length, not just fixed class sizes.

How CIDR works: The Slash Notation (/)

CIDR introduces slash notation (e.g. 192.168.10/24)

In IPv4 address consist of 32 bits (0s and 1s). The number after the slash (/) tells you how many bits are locked for the Network ID

• (leaving the rest of the bits for your independent devices (Hosts))

Total bits in IPv4 = 32.

Host Bits = 32 - CIDR Prefix.

Total Available IPs = $2^{(\text{Host Bits})}$

=> Note: / slash
when you use / (slash) in front of IP
it tells, how big network is!

→ More the slash (28, 30), Network will be smaller.

→ Small the slash number (16, 18),
The more big the Network is.
(Means more IPs exist)